Comparison of Subjective Patient Experiences Between Asleep-Awake-Asleep and Monitored Anesthesia Care Techniques During Awake Craniotomy

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Background: Awake craniotomy (AC) with functional brain mapping is the standard approach for the resection of brain tumors in eloquent areas. Two main anesthetic techniques are commonly used: "Asleep-Awake-Asleep" (SAS) and "Monitored Anesthesia Care" (MAC). SAS involves general anesthesia during nonawake phases, while MAC utilizes conscious sedation, allowing for spontaneous ventilation without invasive airway devices. Both are effective and safe, but information on patient-reported intraoperative experiences is lacking. Here, we compare the subjective experience, postoperative quality of life, and operating room time of patients managed by SAS versus MAC.

Methods: We performed a postoperative telephone survey of all consecutive patients who underwent AC for eloquent brain tumors at the University Medical Center Freiburg, Germany, between October 2018 and April 2024. SAS was used until November 2023, while MAC was used thereafter. The third part of the Beez interview protocol and the EuroQol EQ-5D-5L (including the EQ Visual Analog Scale) were delivered to assess patient-reported intraoperative experience and current health-related quality of life. Patient-reported outcomes and operating room time were compared between the 2 groups.

Results: Thirty-four of 40 patients (17 in each group) were available for telephone interview with more female patients in the SAS group compared with the MAC group (82% vs. 12%, $P < 0.001$). Other baseline parameters were balanced. Patient-reported intraoperative experience and quality of life were similar between the SAS and MAC groups. Operating room times were significantly shorter for MAC (366 ± 107 min) than for SAS (453 ± 81 min) ($P = 0.011$).

Conclusions: Our institutional protocols for SAS and MAC resulted in similar intraoperative experience and postoperative QoL, while MAC procedures had shorter operating room times. Larger studies are required to confirm these findings.

Key Words: anesthesia, awake craniotomy, asleep-awake-asleep, monitored anesthesia care, patient-reported outcomes

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Awake craniotomy (AC) with direct electrical stimulation (DES) is a common approach for removing brain tumors located near eloquent areas, including language, motor, sensory, and cognitive regions.[1–5] AC allows maximal tumor resection while preserving neurological function and quality of life, which may ultimately improve survival.[3,6–8] Despite its benefits, AC is challenging for the medical team and requires active patient participation.[1,2,9] While generally well-tolerated, it demands expertise in airway management, sedation, and pain control, as well as the ability to manage potential complications like seizures, hypoxia, and neurological deficits.[2,3,10,11]

The procedure consists of 3 phases: preawake (positioning and craniotomy), awake (mapping and resection), and postawake (closure).[2,6,11] Two main anesthetic techniques exist: "Asleep-Awake-Asleep" (SAS) and "Monitored Anesthesia Care" (MAC).[12] SAS involves general anesthesia before and after the awake phase, with the usage of invasive airway devices.[1,3,13] In contrast, MAC avoids invasive airway management through conscious sedation with spontaneous ventilation. Both may include a regional scalp nerve block.[3,6,11]

Both techniques are effective; however, limited research has explored patient experiences. Existing studies suggest high satisfaction with both anesthetic approaches, with minimal pain or distress.[1,3,9,11,14–17] To date, limited research has compared patient-reported experiences after SAS and MAC for awake craniotomy.

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METHODS

Study Design and Patients

Ethical approval for this study was obtained from the research ethics committee of the University Medical Center Freiburg on July 18, 2024 (Reference: 24-1275-S1). The primary objective was to compare subjective patient experiences between 2 anesthetic protocols during awake brain surgery. All patients who underwent AC for brain tumors in eloquent areas at the University Medical Center Freiburg, Germany, between October 2018 and April 2024 were retrospectively included in a database. At our institution, SAS was used until November 2023, while MAC was used thereafter. The choice of anesthetic technique was based solely on institutional practice at the time and was not influenced by patient comorbidities, tumor characteristics or expected duration of surgery. Operating room and surgical times were recorded. Operating room time was defined as the interval from the patients' entry into the operating room to their exit. Telephone surveys assessing both intraoperative experience and postoperative health-related quality of life were conducted in May 2024.

Anesthesia and Surgical Procedure

The medical team comprised a neurosurgeon, a neuroanesthesiologist and a neuropsychologist or trained neurosurgical resident for intraoperative testing. Preoperatively, all patients underwent preanesthesia assessment and neuropsychological testing. Preoperative medications included antibiotic prophylaxis with cefuroxime and, if indicated, therapy with dexamethasone or prophylactic antiepileptic treatment with levetiracetam. Patients fasted for 6 hours for solid food and 2 hours for clear liquid before the planned induction of anesthesia. Patients were equipped with standard monitoring consisting of pulse oximetry, electrocardiography (ECG), noninvasive blood pressure measurement (NIBP) and BIS (bispectral index) processed EEG monitoring (Covidien BIS Monitoring System, Medtronic plc, Ireland/USA), and a peripheral intravenous catheter was placed. Then, an arterial line and a second large-diameter intravenous catheter were placed under local anesthesia with lidocaine (MAC group) or after induction of general anesthesia (SAS group).

All patients received paracetamol, metamizole or celecoxib in combination with an antiemetic prophylaxis with dexamethasone and/or ondansetron. After insertion of a urinary catheter, patients were positioned in a semi-lateral/park bench position, facing the anesthesiologist and/or the person for intraoperative testing. Both groups received a regional scalp nerve block with ropivacaine (0.2%, maximum of 60 mL) before pinning the head into a Mayfield skull clamp.

For SAS, patients received general anesthesia in preawake and postawake phases, using total intravenous anesthesia (TIVA). Anesthesia was induced with a target-controlled infusion (TCI) for propofol (Schnider model; target concentration [CeT] 6 to 7 μg/mL), remifentanil (0.2 to 0.3 μg/kg/min) and the airway was secured with a laryngeal mask. Anesthesia was maintained with propofol (CeT between 2.5 and 4 μg/mL) and remifentanil (0.1 to 0,3 μg/kg/min) according to BIS values (target 40 to 60). After dural exposure, the laryngeal mask was removed and patients were awakened. Following brain mapping, sedation was resumed with remifentanil (0.03 to 0.2 μg/kg/min).

In the MAC group, conscious sedation with spontaneous breathing was primarily achieved by a continuous infusion of dexmedetomidine of 0.5 to 1.0 μg/kg/h. In cases where patients complained of pain, remifentanil was also applied in a dose of 0.05 to 0.1 μg/kg/min. When patients asked for more sedation, propofol-TCI with initial CeT at 1 μg/mL was started. The dose was escalated in increments of 0.2 μg/mL to a maximal CeT of 2 μg/mL. Patients received oxygen through high-flow nasal cannulas (Fisher and Paykel Health care). Real-time transcutaneous carbon dioxide ($TcCO_2$) monitoring (Sentec, Switzerland/USA, V-Sign Sensor) and sequential arterial blood gas analyses were performed to prevent hypercapnia by adjusting remifentanil infusion rate.

Surgery was guided by neuronavigation, using a microscope if required. Cortical mapping was conducted using a classic Ojeman stimulator (pulse duration of 1 ms, biphasic wave pulse rate of 60 Hz), with stimulation intensity ranging from 2 to 6 mA. Stimulation was applied to cortical or subcortical areas for 1 to 3 seconds. Intraoperative testing included positive motor and sensory mapping, motor tasks for negative mapping, visual tasks, and speech tasks like counting, word repetition, naming and conversation.

Postoperatively, patients were transferred to the postanesthesia care unit (PACU) and monitored for at least 3 hours. Patients with a pain score ≥ 5 on a 0 to 10 numerical scale received piritramide. Nausea and vomiting were treated with intravenous ondansetron.

Data Collection and Postoperative Survey

Demographic data (age, sex), tumor characteristics (location, hemisphere, pathology, WHO grade, mutations, tumor volume) and surgical details (anesthesia technique, operating room times) were retrospectively collected from patient records. DeepUnity Diagnostics version 2.1.0.0 (Dedalus Group, Italy/Germany) and BrainLab version 7.0 (Brainlab SE, Munich, Germany) were used for the evaluation of imaging data and for tumor volume measurements. Postoperative telephone surveys were conducted in May 2024. These surveys included validated questionnaires as the third part of the Beez interview protocol[16] (SDC 1, Supplemental Digital Content 1, http://links.lww.com/JNA/A876) and the EuroQol 5-dimension 5-level questionnaire (EQ-5D-5L), including the EQ Visual Analog Scale (VAS)[18] (SDC 2, Supplemental Digital Content 2, http://links.lww.com/JNA/A877). The Beez interview protocol comprises 7 items: (1) recollection of the awake surgery, (2) discomfort experienced during surgery and its causes, (3) adequacy of preoperative preparation, (4) fear during surgery, (5) experience of intraoperative testing, (6) perceived duration of the awake

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phase, and (7) suggestions for improvement. In this study, item 7 was omitted. Patient quality of life was assessed with the EuroQol EQ-5D-5L and the EQ Visual Analog Scale. The EQ-5D-5L includes 5 dimensions (5D) addressing mobility, self-care, daily activities, physical complaints, and anxiety/depression. Each question is rated on a 5-level scale (5L), ranging from (a) no problems/restrictions to (e) extreme problems/restrictions. The Visual Analog Scale (VAS) is used to assess the patient's overall health status, with 0% representing the worst imaginable health and 100% representing the best imaginable health.

Statistical Analysis

The statistical analysis was performed using IBM SPSS Statistics Version 29. Quantitative and qualitative variables were expressed as mean values +/− SD or percentages in each category. Graphic representations were made using GraphPad Prism Version 10. Statistical significance was set at a $P < 0.05$. Comparisons of qualitative and quantitative variables were made by means of the Student t test for independent-samples or the Pearson χ^2 test.

RESULTS

We included a total of 40 patients who underwent AC for brain tumors in eloquent areas. Of these, 34 patients were available for a postoperative telephone interview. Six patients (SAS: n = 5; MAC: n = 1; female: n = 4; male: n = 2) were excluded. One was unreachable, one was unable to participate because of aphasia, and 4 died as a result of their underlying brain tumor. Among the nonrespondents, 5 had been diagnosed with glioblastoma WHO grade IV and one with an anaplastic astrocytoma. The SAS and MAC protocols were each used in 17 patients. The mean time from surgery to interview were 882 ± 634 days in the SAS group and 99 ± 36 days in the MAC group.

Demographics, Tumor Characteristics, and Surgical Data

The mean patient age was 41 ± 13 years in the SAS and 45 ± 13 years in the MAC group. In the SAS group, 82% of the patients were female versus 12% in the MAC group ($P < 0.001$); other baseline parameters were balanced (Table 1).

There were no statistically significant differences in tumor characteristics between the 2 groups (Table 1). The majority of brain tumors were located in the left hemisphere. In both groups, most tumors were low-grade gliomas (WHO II graded, IDH-mutated), with more WHO grade IV tumors in the MAC group (29%) than in the SAS group (18%). Neuropathological analysis revealed astrocytoma in 62%, oligodendroglioma in 21%, and glioblastoma in 15%.

Operating room time was significantly shorter for MAC (366 ± 107 min) than for SAS (453 ± 81 min; $P = 0.011$). The mean duration of the surgical procedure was similar between the MAC (208 ± 95 min) and the SAS group (254 ± 71 min; $P = 0.118$).

TABLE 1. Patient and Tumor Characteristics

	SAS (n = 17), n (%)	MAC (n = 17), n (%)	*P*
Age (y): mean ± SD	41.3 ± 13.2	44.9 ± 13.4	0.437[a]
Sex			< 0.001[b]*
Male	3 (17.6)	15 (88.2)	
Female	14 (82.4)	2 (11.8)	
Tumor location			0.703[b]
Frontal	7 (41.2)	3 (17.6)	
Perirolandic	2 (11.8)	3 (17.6)	
Parietal	2 (11.8)	1 (5.9)	
Parieto-occipital	1 (5.9)	2 (11.8)	
Insular involvement	3 (17.6)	5 (29.4)	
Basal ganglia	—	1 (5.9)	
More than 2 brain lobes	2 (11.8)	2 (11.8)	
Hemisphere			0.671[b]
Left	13 (76.5)	14 (82.4)	
Right	4 (23.5)	3 (17.6)	
Pathology			0.085[b]
Astrocytoma	9 (52.9)	12 (70.6)	
Oligodendroglioma	4 (23.5)	3 (17.6)	
Glioblastoma	3 (17.6)	2 (11.8)	
Other	1 (5.9)	—	
WHO grade			0.543[b]
I	—	1 (5.9)	
II	10 (58.8)	8 (47.1)	
III	4 (23.5)	3 (17.6)	
IV	3 (17.6)	5 (29.4)	
IDH-mutated			1.000[b]
Yes	13 (76.5)	13 (76.5)	
No	4 (23.5)	4 (23.5)	
1p/19q-codeleted			0.671[b]
Yes	4 (23.5)	3 (17.6)	
No	13 (76.5)	14 (82.4)	
MGMT-promotor-methylated			0.948[b]
Yes	2 (11.8)	2 (11.8)	
No	15 (88.2)	15 (88.2)	
Tumor volume (cm^3): mean ± SD	16.7 ± 15.9	27.8 ± 22	0.103[a]

Data are expressed as frequency (%) or mean ± SD.
[a]p: t test (comparison of means).
[b]p: χ^2 test (comparison of proportions).
*: Statistical significance ($P < 0.05$).
IDH indicates isocitratdehydrogenase; MAC, monitored anesthesia care; MGMT, O6-methylguanine-DNA-methyltransferase; SAS, asleep-awake-asleep; WHO, World Health Organization.

Postoperative Survey

We found similar responses to the 6 items of the Beez interview protocol as summarized in Figure 1.

Concerning item 1, more than half of patients treated with MAC could remember most details of their AC, whereas the rest could remember only parts. Recollection in SAS was similar with 47% remembering most details of the surgery. Twelve percent of the SAS group could not remember any details.

Intraoperatively, most patients did not report pain, discomfort or anxiety. Discomfort (item 2) was reported by 41% of patients in both the SAS and MAC groups. In the SAS group, discomfort was primarily related to the body position on the operating table and the clamp head fixation, while in MAC, none of the patients reported pain from surgical positioning.

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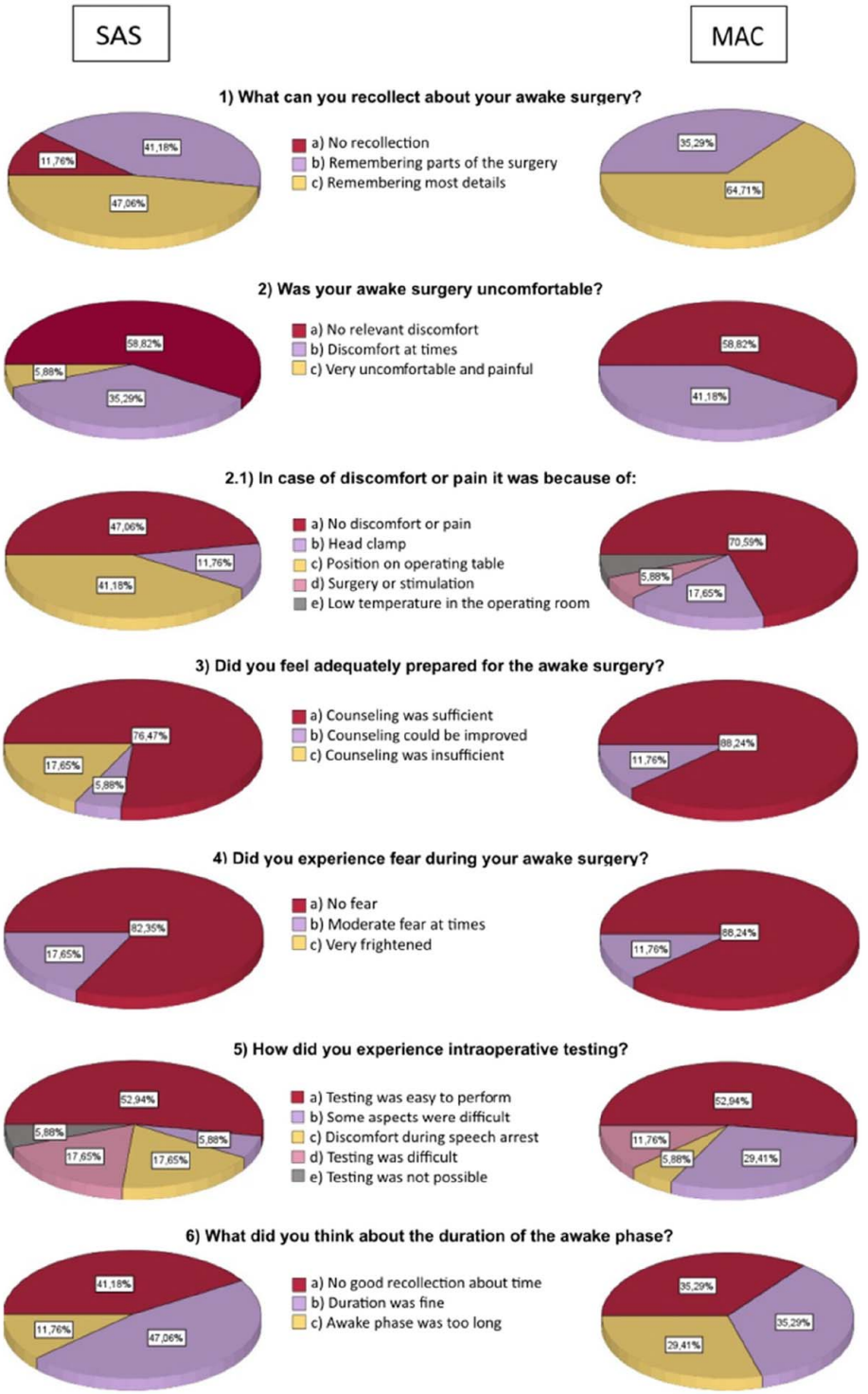


FIGURE 1. Beez interview protocol results. Results of the 6 items from the Beez interview protocol: (1) recollection of the awake surgery, (2) discomfort experienced during surgery and its causes, (3) adequacy of preoperative preparation, (4) fear during surgery, (5) experience of intraoperative testing, and (6) perceived duration of the awake phase. Item 7 (suggestions for improvement), was not included in this study. MAC indicates monitored anesthesia care; SAS, asleep-awake-asleep. full color online

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Seventy-six percent in SAS and 88% in MAC felt adequately prepared for their awake surgery with intraoperative testing (item 3), whereas only 3 patients considered preoperative preparation insufficient.

The majority of patients reported no intraoperative anxiety (item 4) (female: n = 12; male: n = 17), while 5 patients experienced moderate levels of anxiety (female: n = 4; male: n = 1). Notably, none of the patients reported a high level of anxiety.

Intraoperative testing (item 5) was considered easy to perform by 53% of patients in each group, while 6% in SAS and 29% in MAC found some tests difficult. Specifically, 18% in the SAS group and 6% in the MAC group experienced discomfort when electrical stimulation induced speech or motor arrest.

One third of both groups had no precise recollection about time (item 6), although 12% in SAS and 29% in MAC signed that the awake phase was too long.

No significant differences were found in the EuroQol EQ-5D-5L, regarding mobility, self-sufficiency/selfcare, daily activities, physical complaints or anxiety/depression (Fig. 2). The mean reported overall health by VAS in SAS was 78 ± 15%, similar to the MAC group (71% ± 15%) at the time of the postoperative survey (Fig. 3). In SAS, 9 patients had returned to work, while in MAC only 4 patients were working again at the timepoint of the postoperative survey.

DISCUSSION

Awake craniotomy is a highly demanding procedure requiring multidisciplinary teamwork and patient participation. In this study, we found similar subjective patient experience and health-related quality of life between the SAS and MAC anesthetic protocol, although our sample was not balanced in sex distribution and time between surgery and assessment.

Patient comfort and cooperation are paramount to the success of AC and functional testing, and clinical and patient perspectives during surgery should be considered.[2,14] Intraoperative pain management is crucial for good patient cooperation. Our survey showed that most patients did not experience pain or discomfort during surgery, and when it did occur, discomfort was typically related to prolonged immobility in the same body position and clamp head fixation. Interestingly, no patients in the MAC group reported pain due to positioning. Patients undergoing MAC may have actively participated in the positioning phase resulting in greater patient comfort.[1] In our MAC group, we adopted a semilateral positioning, which may have contributed to increased patient satisfaction. These results are consistent with previous work. In a European low-grade glioma network survey, the most frequently reported causes of pain were dural and cerebral vessel manipulation, skull clamp pins and body position.[6] Similar results were reported by Khu et al,[14] where cranial fixation and uncomfortable operative position were the main reasons for discomfort.

In a study of 105 patients undergoing awake glioma surgery, Beez et al[16] reported higher intraoperative anxiety levels among women and patients younger than 60 years. In our cohort, the SAS group had a notably higher proportion of female patients (82%), which may have influenced subjective reported experiences. However, we found similar reported anxiety between sexes and anesthetic techniques, and none of the patients in either group reported high levels of intraoperative anxiety. Anxiety during awake craniotomy appears mostly due to unfamiliar sights, sounds and sensations in the operating room and perception of a lack of information before or during surgery.[9] Wajer et al[19] discussed that anxiety, depression or post-traumatic stress may be related to medical diagnosis and prognosis, rather than to the AC. There are studies suggesting more anxiety when using MAC.[6,17]

In both groups, the majority of patients reported a high degree of recollection during surgery and that the duration of the awake phase was adequate, similar to previous studies.[14] Most of the patients stated that preoperative counseling was adequate, which may have reduced anxiety and regulated expectations.[2,14] Moreover, it builds trust in the surgeon and medical team.[14] In a review by Khu and colleagues, most patients were satisfied with their AC, similar to our study. Participating in their surgery gave them a sense of control over their disease.[14]

We found no significant differences in health-related quality of life, assessed with the EuroQol EQ-5D-5L (including the EQ VAS). In a case series of 14 glioma patients undergoing awake craniotomy, Tan et al[20] demonstrated that higher preoperative depression and stress scores may be associated with worse health-related quality of life. The reason for fewer patients resuming work in our MAC group might be that some of them were still in rehabilitation or follow-up treatment, as their surgery was more recent than in the SAS group.

Previous studies comparing SAS and MAC from an anesthesiological point of view found none of them superior to the other. A review by Stevanovic et al[3] showed similar rates of AC failure, seizures, and neurological deficits, though unplanned conversions to general anesthesia occurred more often with SAS. SAS may offer better pain control and prevent accidental patient movement, while controlled ventilation can reduce brain swelling through hyperventilation.[2] However, it's linked to a higher incidence of hypertension during awakening, increasing intraoperative bleeding risk.[2,6] Furthermore, airway stimulation may raise intracranial pressure.[2] In contrast, MAC avoids potential airway complications of general anesthesia. Paños et al[1] reported complications like tachycardia and desaturation only in SAS patients. However, balanced sedation is key in MAC to avoid respiratory depression.[12] Higher $PaCO_2$ levels—associated with brain swelling—have been seen with MAC.[2,4] In our MAC group, nasal high-flow therapy, $TcCO_2$ monitoring, and arterial blood gas analyses helped maintain stable $PaCO_2$ levels during spontaneous breathing.[21]

In this study, the MAC group had significantly shorter operating room durations, consistent with previous literature.[1,4,11,17,22] Intubation, extubation, and waiting for full awakening before testing during the SAS technique may prolong the procedure.

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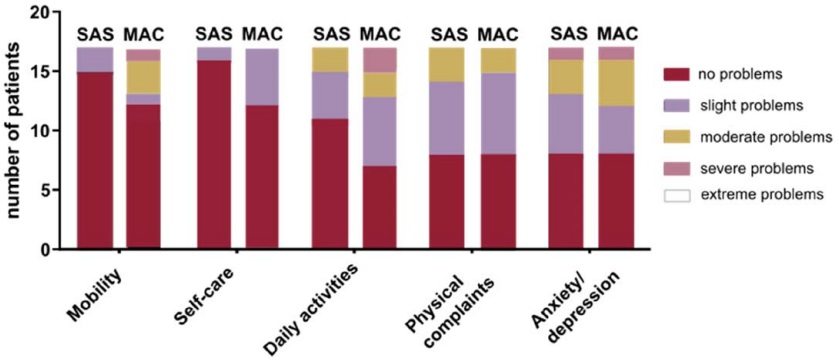


FIGURE 2. EuroQol EQ-5D-5L results. The 5 dimensions (5D) assessed were mobility, self-care, daily activities, physical complaints, and anxiety/depression. Each was rated on a 5-level scale (5L) from no to extreme problems. MAC indicates monitored anesthesia care; SAS, asleep-awake-asleep. full color online

Our study has limitations. First, we included a relatively small sample size from a single center, which limits the statistical power and generalizability. In addition, there were group imbalances, particularly in sex distribution, with a higher proportion of female patients in the SAS group. The reason for this disparity remains unclear. However, it is unlikely to reflect selection bias, as the allocation of anesthetic technique was based solely on the time period during the procedure was performed, with all patients included retrospectively. Nonetheless, the presence of such demographic imbalances underscores the need for larger, prospectively designed studies to ensure more robust conclusions. Moreover, clinical differences or comorbidities beyond anesthetic technique may have varied over the study period, potentially influencing results. The time interval between surgery and the follow-up interview was significantly longer in the SAS than in the MAC group, which increases the risk of recall bias between the groups, as patients' memories of sensations and discomfort may fade with time. In this study population, cognitive decline associated with progressive brain disease may further compromise recall, warranting caution in interpreting our findings.

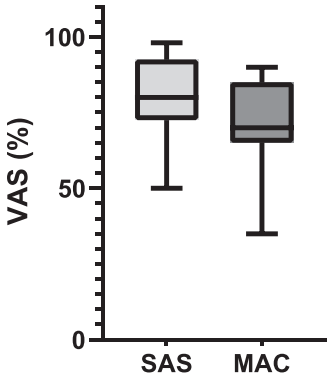


FIGURE 3. EQ Visual Analog Scale (VAS). The EQ VAS was used to assess patient's subjective overall health status, with 0% representing the worst imaginable health and 100% the best imaginable health. Boxplots show the mean VAS score (in percent) within each box, while the whiskers indicate the minimum and maximum values. MAC indicates monitored anesthesia care; SAS, asleep-awake-asleep.

In conclusion, this exploratory study suggests that the majority of patients undergoing AC for brain tumor resection reported a generally positive experience and high levels of satisfaction, regardless of the anesthetic technique used. Subjective experiences, including intraoperative pain, discomfort and anxiety, appeared broadly comparable across the cohort. These findings indicate that successful AC, as measured by patient-reported metrics, may be achievable with both SAS and MAC; though larger, prospective studies are warranted to validate these findings.

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